

Course 3371

3 Credits: 3

Program Core

Source-grounded

Introduction to IC Fabrication

- To introduce the fundamentals of IC fabrication process.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3371 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3371.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	3371
Course title	Introduction to IC Fabrication
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:3 T:0 P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the fundamentals of IC fabrication process.

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding preparation and wafer cleaning.	See official syllabus
CO2	Explain the process of epitaxy and oxidation. 11 Understanding	See official syllabus
CO3	Explain lithography and etching process. 10 Understanding Illustrate the process of deposition, diffusion, ion	See official syllabus
CO4	12 Understanding implantation and packaging. Series Exam 2	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	cleaning	4	As specified
Module 2	Explain the process of Epitaxy and Oxidation.	4	As specified
Module 3	Explain Lithography and Etching process.	4	As specified
Module 4	Packaging.	4	As specified

MODULE 1

cleaning

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cleaning
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Understanding fabrication

2 Understanding fabrication is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding fabrication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding fabrication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding fabrication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding fabrication ഫാബ്രിക്കേഷൻ
definition/meaning ഫാബ്രിക്കേഷൻ. ഫാബ്രിക്കേഷൻ ഫോളോവിംഗ്, steps,
application ഫാബ്രിക്കേഷൻ ഫോളോവിംഗ്. Technical terms English- ഫാബ്രിക്കേഷൻ
ഫോളോവിംഗ്.

Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing

Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the methods of single crystal growth 3 understanding explain chemical-mechanical polishing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓരോ Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing ഫാബ്രിക്കേഷൻ ഫ്രോസസ്സ് definition/meaning ഫാബ്രിക്കേഷൻ ഫ്രോസസ്സ്. ഫാബ്രിക്കേഷൻ ഫ്രോസസ്സ്, steps, application ഫാബ്രിക്കേഷൻ ഫ്രോസസ്സ് ഫാബ്രിക്കേഷൻ. Technical terms English-ഓരോ ഫാബ്രിക്കേഷൻ ഫ്രോസസ്സ്.

2 Understanding technique

2 Understanding technique is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding technique using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding technique should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding technique means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding technique
 definition/meaning .
 application
 Technical terms English-
 .

Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.

Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain wafer preparation 3 understanding crystal growth, wafer preparation: introduction & historical perspective- clean room concept- growth of single crystal si-czochralski(cz) and float zone(fz) processes. surface contamination-chemical-mechanical polishing. wafer preparation-di water, chemical cleaning-rca, processing consideration- getting the thermal stress factor etc. should be discussed with attention

to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. definition/meaning. steps, application. Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

cleaning		
M1.01	Explain the concept of clean room in IC	2
Understanding		
	fabrication	
M1.02	Explain the methods of single crystal growth	3
Understanding		
M1.03	Explain chemical-mechanical polishing	2
Understanding		
	technique	
M1.04	Explain wafer preparation	3
Understanding		
Contents:		
Crystal growth, wafer preparation: Introduction & historical perspective- Clean room		
concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes.		
Surface		
Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical		
cleaning-RCA, processing consideration- getting the thermal stress factor etc.		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the process of Epitaxy and Oxidation.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the process of Epitaxy and Oxidation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth

Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the process of epitaxy 3 understanding illustrate epitaxial valuation and growth should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എപിറ്റാക്സി ഉപയോഗിച്ച് ഉപരിതലത്തിൽ പാളം വളർത്തുന്നതിന്റെ പ്രക്രിയ വിശദീകരിക്കുക. 3 Understanding
Illustrate Epitaxial valuation and growth എപിറ്റാക്സിയുടെ നിർവ്വചനം/അർത്ഥം
എപിറ്റാക്സി എന്നതിന്റെ അർത്ഥം, steps, application എപിറ്റാക്സി ഉപയോഗിച്ച്
എപിറ്റാക്സി ഉപയോഗിച്ച്. Technical terms English-എപിറ്റാക്സി ഉപയോഗിച്ച്.

3 Understanding mechanism

3 Understanding mechanism is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mechanism means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding mechanism ഏകദേശം
definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം, steps,
application ഏകദേശം ഏകദേശം. Technical terms English-ഏകദേശം
ഏകദേശം.

Outline the process of oxidation

Outline the process of oxidation is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the process of oxidation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the process of oxidation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the process of oxidation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓക്സീകരണം Outline the process of oxidation ഓക്സീകരണം
ഓക്സീകരണം definition/meaning ഓക്സീകരണം. ഓക്സീകരണം ഓക്സീകരണം, steps,
application ഓക്സീകരണം ഓക്സീകരണം. Technical terms English-ഓക്സീകരണം
ഓക്സീകരണം.

Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.

Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the kinetics of oxidation 2 understanding epitaxy and oxidation: epitaxy process-vapor phase epitaxy, molecular beam epitaxy. epitaxial valuation-growth mechanism and kinetics. oxidation techniques-dry and wet oxidation. device isolation-locos-kinetics of oxidation. deal grove's model-oxidation system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system. definition/meaning . , steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Explain the process of Epitaxy and Oxidation.

M2.01	Explain the process of Epitaxy	3
Understanding		

	Illustrate Epitaxial valuation and growth	
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M2.02		3
Understanding		

	mechanism	
--	-----------	--

M2.03	Outline the process of oxidation	3
Understanding		

M2.04	Describe the kinetics of oxidation	2
Understanding		

	Series Test-1	1
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Contents:

Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy.
Epitaxial Valuation-Growth mechanism and kinetics.

Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of
Oxidation.

Deal Grove's Model-Oxidation system.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain Lithography and Etching process.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Lithography and Etching process.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the concept of Lithography

Illustrate the concept of Lithography is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concept of Lithography using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concept of lithography should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concept of Lithography means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോ ഉദാഹരണവും Lithography
എന്താണിത് definition/meaning എന്താണിത്. എന്താണിത് എന്താണിത്,
steps, application എന്താണിത് എന്താണിത് എന്താണിത്. Technical terms English-ഓരോ
എന്താണിത്.

Explain different types of Lithographic processes

Explain different types of Lithographic processes is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different types of Lithographic processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of lithographic processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different types of Lithographic processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain different types of Lithographic processes. Define/meaning. Steps, application. Technical terms English-Malayalam.

Explain the concept of etching

Explain the concept of etching is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of etching using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of etching should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of etching means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain the concept of etching
 definition/meaning .
 application
 Technical terms English-
 .

Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and

Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the different types of etching process 3 understanding lithography & etching: lithography-overview of lithography-radiation sources-masks-

photoresist-components of photoresist- optical aligners-resolution-depth of focus-advanced lithography-electron beam lithography-x-ray lithography-ion lithography. etching-anisotropy-selectivity-wet etching, plasma etching, plasma properties-reactive - ion etching. illustrate the process of deposition, diffusion, ion implantation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു പറയുന്ന വിവിധ തരം എ칭ിംഗ് പ്രക്രിയകളെക്കുറിച്ച് 3 Understanding Lithography & Etching: Lithography-Overview of Lithography- Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners- Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and താഴെ പറയുന്നവയെക്കുറിച്ച് നിങ്ങളുടെ നിർവ്വചനം/അർത്ഥം എഴുതുക. താഴെ പറയുന്നവയെക്കുറിച്ച്, steps, application താഴെ പറയുന്നവയെക്കുറിച്ച് എഴുതുക. Technical terms English-ൽ എഴുതുക താഴെ പറയുന്നവയെക്കുറിച്ച്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Explain Lithography and Etching process.

M3.01	Illustrate the concept of Lithography	2
Understanding		

M3.02	Explain different types of Lithographic processes	4
Understanding		

M3.03	Explain the concept of etching	1
Understanding		

M3.04	Explain the different types of etching process	3
Understanding		

Contents:

Lithography & Etching:

Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography- Electron

beam Lithography-X-ray Lithography-Ion Lithography.

Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties- Reactive -

Ion Etching.

Illustrate the process of Deposition, Diffusion, Ion implantation and

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Packaging.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Packaging.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain Diffusion mechanism

Explain Diffusion mechanism is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Diffusion mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain diffusion mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Diffusion mechanism means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Explain Diffusion mechanism
definition/meaning .
steps,
application . Technical terms English-
.

Explain various ion implantation techniques

Explain various ion implantation techniques is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various ion implantation techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various ion implantation techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various ion implantation techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഈ Explain various ion implantation techniques
ഈ definition/meaning ഈ. ഈ ഈ, steps, application ഈ. Technical terms English-ഈ ഈ.

Explain different deposition mechanisms

Explain different deposition mechanisms is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different deposition mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different deposition mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different deposition mechanisms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓം Explain different deposition mechanisms
ഓംഓംഓംഓംഓം ഓംഓം definition/meaning ഓംഓംഓംഓംഓംഓം. ഓംഓംഓം ഓംഓംഓം ഓംഓംഓം,
steps, application ഓംഓംഓം ഓംഓംഓംഓം ഓംഓംഓം. Technical terms English-ഓം ഓംഓംഓം
ഓംഓംഓംഓംഓം.

Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in>

Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning- Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain metallization and packaging 3 understanding deposition, diffusion, ion implantation and packaging: model of diffusion in solids- atomic diffusion mechanism-measurement technique-range theory-implantation equipment-annealing shallow junction-high energy implantation-deposition process- chemical vapor deposition-physical vapor deposition-patterning-metallization and packaging-baked ic design. text / reference: t/r book title/author t1 s.m sze, “vlsi technology”, mcgraw hill publication, 2003. t2 s.k gandhi, “vlsi fabrication principle”, willy-india pvt ltd, 2008. t3 suresh babu “solid state device & technology”, sanguine, 2005. r1 j.d plummer, m.d deal and peter b. griffin, “silicon vlsi technology: fundamentals, practice and modelling”, pearson education publication, 2009. r2 stephen a. campbell, “fabrication engineering at the micro and nano scale”, oxford university press, 2013 online resources: sl. no. website link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-

Aspect	What to prepare
	Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in> ഫലപ്രദമായ രീതിയിൽ definition/meaning നൽകുക. ഏതെങ്കിലും രീതിയിൽ, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Packaging.

M4.01	Explain Diffusion mechanism Understanding	3
M4.02	Explain various ion implantation techniques Understanding	3
M4.03	Explain different deposition mechanisms Understanding	3
M4.04	Explain metallization and packaging Understanding	3
	Series Test-II	1

Contents:

Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids-

Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design.

Text / Reference:

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding fabrication. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding fabrication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോ ചോദ്യത്തിനും നിങ്ങളുടെ ഉത്തരം definition, principle/steps, application എന്നിവ ഉൾക്കൊള്ളിക്കണം.

Define or explain Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോ ചോദ്യത്തിനും നിങ്ങളുടെ ഉത്തരം definition, principle/steps, application എന്നിവ ഉൾക്കൊള്ളിക്കണം.

Define or explain 2 Understanding technique. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding technique*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഓരോ ചോദ്യത്തിനും നിങ്ങളുടെ ഉത്തരം definition, principle/steps, application എന്നിവ ഉൾക്കൊള്ളിക്കണം.

Define or explain Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress*

factor etc.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding mechanism. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding mechanism*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the process of oxidation. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the process of oxidation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Illustrate the concept of Lithography. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the concept of Lithography.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain different types of Lithographic processes. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different types of Lithographic processes.* State the governing principle, list the key components or

stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the concept of etching. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the concept of etching*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the different types of etching process 3

Understanding Lithography & Etching: Lithography-Overview of Lithography- Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy- Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Explain Diffusion mechanism. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Diffusion mechanism*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain various ion implantation techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Explain various ion implantation techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain different deposition mechanisms. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different deposition mechanisms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory- Implantation equipment-Annealing shallow junction-High energy implantation- Deposition process- Chemical vapor deposition-Physical vapor deposition-

Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in>. (3 marks)

Answer: Begin with the standard definition or identity of *Explain metallization and packaging* 3 *Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. <http://www.asic-world.com> 2. <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> 3. <https://www.testbench.in>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding fabrication.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the concept of Lithography.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain Diffusion mechanism.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link <http://www.asic-world.com>
- <https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech> <https://www.testbench.in>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 3371. Verify institutional updates against the current official curriculum.